gpu-mode / resource-stream

Code Issues Pull requests Actions Projects Security Insights

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GPU programming related news and material links

discord.gg/gpumode

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msaroufim Merge pull request #23 from gpu-mode/codex/add-flaggems-to-triton-ker...

95a4f68 · 3 weeks ago

.gitignore	Initial commit	3 years ago
LICENSE	CUDA MODE -> GPU MODE	2 years ago
README.md	docs: add FlagGems Triton resource	3 weeks ago

README MIT license

GPU MODE Resource Stream

GPU MODE 26180 members

https://discord.gg/gpumode

Here you find a collection of CUDA related material (books, papers, blog-post, youtube videos, tweets, implementations etc.). We also collect information to higher level tools for performance optimization and kernel development like Triton and `torch.compile()` ... whatever makes the GPUs go brrrr.

You know a great resource we should add? Please see How to contribute.

Lectures / Reading Group Live Sessions

You find a list of upcoming lectures in the Events option in the channel list (side bar) of our discord server.

Recordings of the weekly lectures are published on our YouTube channel. Material (code, slides) for the individual lectures can be found in the lectures repository.

1st Contact with CUDA

- An Easy Introduction to CUDA C and C++
- An Even Easier Introduction to CUDA
- CUDA Toolkit Documentation
- Accelerated Computing Hub NVIDIA curated collection of open source educational resources for general purpose GPU programming.
- Basic terminology: Thread block, Warp, Streaming Multiprocessor: Wiki: Thread Block, A tour of CUDA
- GPU Performance Background User's Guide
- OLCF NVIDIA CUDA Training Series, talk recordings can be found under the presentation footer for each lecture; exercises
- GTC 2022 - CUDA: New Features and Beyond - Stephen Jones
- Intro video: Writing Code That Runs FAST on a GPU
- 12 hrs CUDA tutorial: Introduction of CUDA and writing kernels in CUDA

2nd Contact

- CUDA Refresher

Hazy Research

The MLSys-oriented research group at Stanford led by Chris Re, with alumni Tri Dao, Dan Fu, and many others. A goldmine.

- Building Blocks for AI Systems: Their collection of resources similar to this one, many great links
- Data-Centric AI: An older such collection
- Blog
- ThunderKittens: (May 2024) A DSL within CUDA, this blog post has good background on getting good H100 performance
- Systems for Foundation Models, and Foundation Models for Systems: Chris Re's keynote from NeurIPS Dec 2023

Papers, Case Studies

- A Case Study in CUDA Kernel Fusion: Implementing FlashAttention-2 on NVIDIA Hopper Architecture using the CUTLASS Library
- How to Optimize a CUDA Matmul Kernel for cuBLAS-like Performance: a Worklog
- Anatomy of high-performance matrix multiplication

Books

- Programming Massively Parallel Processors: A Hands-on Approach
- Cuda by Example: An Introduction to General-Purpose Gpu Programming; code
- The CUDA Handbook
- The Book of Shaders guide through the abstract and complex universe of Fragment Shader (not cuda but GPU related)
- Art of HPC 4 books on HPC more generally, does not specifically cover GPUs but lessons broadly apply

Cuda Courses

- HetSys: Programming Heterogeneous Computing Systems with GPUs and other Accelerators
- Heterogeneous Parallel Programming Class (YouTube playlist) Prof. Wen-mei Hwu, University of Illinois
- Official YouTube channel for "Programming Massively Parallel Processors: A Hands-on Approach", course playlist: Applied Parallel Programming
- Programming Parallel Computers; covers both CUDA and CPU-parallelism. Use Open Course Version and you can even submit your own solutions to the exercises for testing and benchmarking.

CUDA Grandmasters

Tri Dao

- x: @tri_dao, gh: tridao
- Dao-AILab/flash-attention, paper
- state-spaces/mamba, paper: Mamba: Linear-Time Sequence Modeling with Selective State Spaces, minimal impl: mamba-minimal

Tim Dettmers

- x: @Tim_Dettmers, gh: TimDettmers
- TimDettmers/bitsandbytes, docs: docs
- QLoRA: Efficient Finetuning of Quantized LLMs

Sasha Rush

- x: @srush_nlp, gh: srush
- Sasha Rush's GPU Puzzles, dshah3's CUDA C++ version & walkthrough video
- Mamba: The Hard Way, code: srush/annotated-mamba

Practice

- Adnan Aziz and Anupam Bhatnagar GPU Puzzlers

PyTorch Performance Optimization

- Accelerating Generative AI with PyTorch: Segment Anything, Fast
- Accelerating Generative AI with PyTorch II: GPT, Fast
- Speed, Python: Pick Two. How CUDA Graphs Enable Fast Python Code for Deep Learning
- Performance Debugging of Production PyTorch Models at Meta

PyTorch Internals & Debugging

- TorchDynamo Deep Dive
- PyTorch Compiler Troubleshooting
- PyTorch internals
- Pytorch 2 internals
- Understanding GPU memory: 1: Visualizing All Allocations over Time, 2: Finding and Removing Reference Cycles
- Debugging memory using snapshots: Debugging PyTorch memory use with snapshots
- CUDA caching allocaator: https://zdevito.github.io/2022/08/04/cuda-caching-allocator.html
- Trace Analyzer: PyTorch Trace Analysis for the Masses
- Holistic Trace Analysis (HTA), gh: facebookresearch/HolisticTraceAnalysis

Code / Libs

- NVIDIA/cutlass

Essentials

- Triton compiler tutorials
- CUDA C++ Programming Guide
- PyTorch: Custom C++ and CUDA Extensions, Code: pytorch/extension-cpp
- PyTorch C++ API
- pybind11 documentation
- NVIDIA Tensor Core Programming
- GPU Programming: When, Why and How?
- How GPU Computing Works | GTC 2021 (more basic than the 2022 version)
- How CUDA Programming Works | GTC 2022
- CUDA Kernel optimization Part 1 Part 2
- PTX and ISA Programming Guide (V8.3)
- Compiler Explorer: Inspect PTX: div 256 -> shr 8 example

Profiling

- Nsight Compute Profiling Guide

- mcarilli/nsight.sh - Favorite nsight systems profiling commands for PyTorch scripts
- Profiling GPU Applications with Nsight Systems

Python GPU Computing

- PyTorch
- Trtion, github: openai/triton
- numba @cuda.jit
- Apache TVM
- JAX Pallas
- CuPy NumPy compatible GPU Computing
- NVidia Fuser
- Codon @gpu.kernel, github: exaloop/codon
- Mojo (part of commercial MAX Plattform by Modular)
- NVIDIA Python Bindings: CUDA Python (calling NVRTC to compile kernels, malloc, copy, launching kernels, ..), cuDNN FrontEnd(FE) API, CUTLASS Python Interface
- NVIDIA tilus

Advanced Topics, Research, Compilers

- TACO: The Tensor Algebra Compiler, gh: tensor-compiler/taco
- Mosaic compiler C++ DSL for sparse and dense tensors algebra (built on top of TACO), paper, presentation

News

- SemiAnalysis

Technical Blog Posts

- Cooperative Groups: Flexible CUDA Thread Programming (Oct 04, 2017)
- A friendly introduction to machine learning compilers and optimizers (Sep 7, 2021)

Hardware Architecture

- NVIDIA H100 Whitepaper
- NVIDIA GH200 Whitepaper
- AMD CDNA 3 Whitepaper
- AMD MI300X Data Sheet
- Video: Can SRAM Keep Shrinking? (by Asianometry)

GPU-MODE Community Projects

ring-attention

- see our ring-attention repo

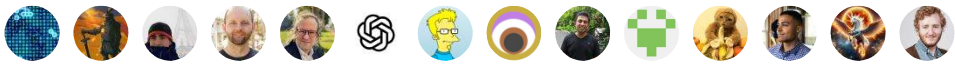
pscan

- GPU Gems: Parallel Prefix Sum (Scan) with CUDA, PDF version (2007), impl: stack overflow, nicer impl: mattdean1/cuda
- Accelerating Reduction and Scan Using Tensor Core Units
- Thrust: Prefix Sums, Reference: scan variants
- CUB, part of cccl: NVIDIA/cccl/tree/main/cub
- SAM Algorithm: Higher-Order and Tuple-Based Massively-Parallel Prefix Sums (licensed for non commercial use only)
- CUB Algorithm: Single-pass Parallel Prefix Scan with Decoupled Look-back
- Group Experiments: johnryan465/pscan, andreaskoepf/pscan_kernel

Triton Kernels / Examples

- `unsloth` that implements custom kernels in Triton for faster QLoRA training
- FlagGems - High-performance Triton operator library (230+ kernels) focused on LLM training and inference
- Custom implementation of relative position attention (link)
- Tri Dao's Triton implementation of Flash Attention: flash_attn_triton.py
- YouTube playlist: Triton Conference 2023
- LightLLM with different triton kernels for different LLMs

Contributors 20



+ 6 contributors